Layer	Technology / Tool	Specific Role in MilestonePay	Hackathon Track / Rule Satisfied
App Framework	Next.js 15 (App Router) + TypeScript + Tailwind CSS	Hosts both the single-screen React frontend and the /api/* backend routes in one clean repository.	Fast full-stack deployment; clean TypeScript typing.
AI & Agent Layer	Gemini 2.5 Flash (@google/genai) + @paypal/agent-toolkit	Evaluates GitHub PR diffs using structured JSON output and invokes PayPal invoicing and order tools.	Grand Prize ($12k) & Best Use of Agentic Commerce ($5k).
PayPal Core APIs	PayPal Invoicing API, Orders/Capture API, Payouts API, Webhooks	Handles incoming client escrow funding, outgoing contractor payouts, and real-time payment state notifications.	Technological Implementation (the official tie-breaker criterion).
API Code Generation	APIMatic PayPal Context Plugin (npx context-plugins install paypal)	Provides verified OpenAPI schemas and TypeScript SDK patterns inside your IDE for Payouts and Webhook signature verification.	Best Use of APIMatic ($1,000 x 3).
Timeline UI	Bryntum Gantt (@bryntum/gantt-react)	Renders the dependency graph, custom payment-status task bars, and automated critical-path date shifting.	Best Use of Bryntum ($1,000 x 3).
Financial Ledger UI	AG Grid Enterprise (ag-grid-react + ag-grid-enterprise)	Renders the escrow ledger with Master/Detail expandable webhook/audit drawers, custom status badges, and pinned summary rows.	Best Use of AG Grid ($5,000 / $2,000 / $1,000 x 3).
State Sync	Zustand (Client) + Server-Sent Events (SSE) or Polling	Keeps the Bryntum Gantt chart and AG Grid rows 100% synchronized from a single milestones[] store.	Design & Presentation rubric criteria.
Cloud Hosting	Render (render.com)	Runs the live Node.js web service so external PayPal Sandbox webhooks can hit a public HTTPS URL 24/7.	Render Sponsor Pool ($1,000 credits).